

Exercise 10.14

考虑两个 IVPs $\begin{cases} v' = f(v, t) \\ v(a) = v_0 \end{cases}$ 和 $\begin{cases} w' = f(w, t) \\ w(a) = 0 \end{cases}$, 显然 IVPs 解分别为 $v(t) = v_0 + \int_a^t f(v, s)ds$ 和 $w(t) = w_0 + \int_a^t f(w(s), s)ds$.

$$\begin{aligned} \|v(t) - w(t)\| &= \|v_0 - w_0\| + \left\| \int_a^t f(v, s) - f(w(s), s)ds \right\| \\ &\leq \|v_0 - w_0\| + \int_a^t \|f(v, s) - f(w(s), s)\|ds \\ &\leq \|v_0 - w_0\| + \int_a^t L\|v(s) - w(s)\|ds \end{aligned}$$

由 Gronwall 不等式,

$$\Rightarrow \|v(t) - w(t)\| \leq \|v_0 - w_0\| \cdot \exp\left\{\int_a^t Lds\right\} = \|v_0 - w_0\| \cdot \exp\{L(t-a)\}.$$

Exercise 10.64

Trapezoidal:

$$C_0 = 0, C_1 = 0, C_2 = -1, C_3 = -\frac{1}{12}, C_4 = -\frac{1}{24}$$

Midpoint:

$$C_0 = 0, C_1 = 0, C_2 = 0, C_3 = \frac{1}{3}, C_4 = \frac{1}{3}$$

Exercise 10.66

为了达到 3 阶精度, 需令 $C_0 = C_1 = C_2 = 0$, 即

$$\sum_{n=0}^s a_n = 0, \quad \sum_{n=0}^s na_n = \sum_{n=0}^s b_n, \quad \sum_{n=0}^s n^2 a_n = \sum_{n=0}^s nb_n$$

. 即 $\rho(1) = 0, \rho'(1) = \sigma(1), \rho'(1) + \rho''(1) = \frac{1}{2}\sigma'(1)$.

Exercise 10.67

Adams-Bashforth 方法:

s 步显式方法, 系数通过积分拉格朗日插值多项式获得。

Listing 1: MATLAB 示例代码

```

1 syms theta;
2 s = input('Enter the value of s: ');
3 theta_nodes = 0 : -1 : -(s-1);
4 beta = zeros(1, s);
5
6 for i = 0 : s-1
7     current_theta = theta_nodes(i+1);
8     Li = 1;
9     for k = 0 : s-1
10         if k ~= i

```

```

11         Li = Li * (theta - theta_nodes(k+1)) / (current_theta - theta_nodes(k
12             +1));
13     end
14     beta(i+1) = int(Li, theta, 0, 1);
15 end
16
17 disp('Coefficients (from beta_s-1 to beta_0):');
18 disp(vpa(beta, 4));

```

Coefficients: [3/2, -1/2]

Coefficients: [23/12, -4/3, 5/12]

Coefficients: [55/24, -59/24, 37/24, -3/8]

Adams-Moulton:

隐式多步法，其系数通过积分拉格朗日插值多项式得到。其插值节点包含当前时刻 t_{n+1} 。

```

1  syms theta;
2  s = input('Enter the value of s (Adams-Moulton): ');
3  theta_nodes = 0 : -1 : -(s-1);
4  beta = zeros(1, s);
5
6  for i = 0 : s-1
7      current_theta = theta_nodes(i+1);
8      Li = 1;
9      for k = 0 : s-1
10         if k ~= i
11             Li = Li * (theta - theta_nodes(k+1)) / (current_theta - theta_nodes(k
12                 +1));
13         end
14     end
15     beta(i+1) = int(Li, theta, 0, 1);
16 end
17 disp('Adams-Moulton Coefficients (\beta_s 到 \beta_0):');
18 disp(vpa(beta, 4));

```

Coefficients: [1/2, 1/2]

Coefficients: [5/12, 8/12, -1/12]

Coefficients: [9/24, 19/24, -5/24, 1/24]

BDF:

隐式方法，通过多项式插值并强制导数条件确定系数。

```

1  syms h t;
2  s = input('Enter the value of s (BDF): ');
3  a = sym('a', [1, s+1]);
4  p = sum(a .* t.^(0:s));

```

```

5
6 eqns = [];
7 for k = 1:s
8     eqns = [eqns, subs(p, t, -k*h) == 0];
9 end
10
11 dpdt = diff(p, t);
12 eqns = [eqns, subs(dpdt, t, 0) == 1];
13
14 [A, B] = equationsToMatrix(eqns, a);
15 sol = linsolve(A, B);
16
17 alpha = [1; -double(sol(2:end))];
18 beta = 1;
19
20 disp(vpa(alpha.', 4));
21 disp(vpa(beta, 4));

```

$$\alpha : [1, -1], \beta : 1$$

$$\alpha : [1, -4/3, 1/3], \beta : 2/3$$

$$\alpha : [1, -18/11, 9/11, -2/11], \beta : 6/11$$

Exercise 10.72

$$\rho(z) = \sum_{j=0}^3 \alpha_j z^j = Z^3 - \frac{18}{11} z^2 + \frac{9}{11} z - \frac{2}{11},$$

$$\sigma(z) = \sum_{j=0}^3 \beta_j z^j = \frac{6}{11} z^3,$$

$$g(z) = \frac{\rho(z)}{\sigma(z)} = \frac{11}{6} - \frac{3}{z} + \frac{3}{2z^2} - \frac{1}{3z^3}.$$

$$g(1) = 0, g'(1) = -1, g''(1) = 2, g^{(4)}(1) = -12.$$

在 $z=1$ 处展开:

$$g(z) = (z-1) - \frac{1}{2}(z-1)^2 + \frac{1}{3}(z-1)^3 - \frac{1}{4}(z-1)^4 + \Theta(-\frac{1}{4}(z-1)^4).$$

Exercise 10.73

$$\text{由 (10.44) } L_u(t_n) = C_0 u(t_n) + C_1 k u_t(t_n) + C_2 k^2 u_{tt}(t_n) + \dots$$

必要性:

假设 LMM 有 p 阶精度, $\forall j = 0, 1, \dots, p$ 有 $C_j = 0, C_{p+1} \neq 0$. 若 $q(t)$ 阶数小于 p , 则 $\forall j = p, p+1, \dots$ 有 $\frac{\partial^j}{\partial t^j} q(t) = \frac{\partial^{j+1}}{\partial t^{j+1}} u = 0$. 此时 $L_u(t_n) = 0$, 又因为已知初始条件精确, 所以解完全精确。

充分性:

$$L_u(t_n) = C_0 u(t_n) + \sum_{j=1}^{\infty} C_j k^j q^{(j-1)}(t_n)$$

因为 $\forall q(t)$ 是 $p-1$ 阶多项式, 所以 $L_u(t_n) = 0$,

$$\text{取 } q(t) = 0 \Rightarrow C_0 = 0, \quad q(t) = t \Rightarrow C_1 = 0, \dots, q(t) = t^{p-1} \Rightarrow C_p = 0.$$

当 q 阶数大于 $p-1$ 时, 不妨设 $q(t) = t^p$, 此时 $L_u(t_n) \neq 0$, 即 $C_{p+1} \neq 0$, 则 $L_u(t_n)$ 为 p 阶精度。

Exercise 10.82

$$\text{当 } n = 0, 1, \dots, s-1 \quad \begin{pmatrix} y_{s-1} \\ y_{s-2} \\ \vdots \\ y_1 \\ y_0 \end{pmatrix} = \begin{pmatrix} 1 & \theta_1 & \theta_2 & \cdots & \theta_{s-1} \\ 0 & 1 & \theta_1 & \cdots & \theta_{s-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & \theta_1 \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{y}_{s-1} \\ \hat{y}_{s-2} \\ \vdots \\ \hat{y}_1 \\ \hat{y}_0 \end{pmatrix}$$

即 $y_n = \sum_{i=0}^{s-1} \theta_{n-i} \hat{y}_i$.

当 $n > s$ 时, 归纳假设 (10.62) 对 $\forall n = m + s - 1$ 成立, 下证对 $n = m + s$ 成立。由已知条件 $y_{n+s} = \phi_{n+s} - \sum_{i=0}^{s-1} \alpha_i y_{n+i}$, $\theta_{n+s} + \sum_{i=0}^{s-1} \alpha_i \theta_{n+i} = 0$. 由归纳假设 $y_{n+i} = \sum_{j=0}^{s-1} \theta_{n+i-j} \hat{y}_j + \sum_{j=s}^{n+i} \theta_{n+i-j} \hat{\phi}_j$,

$$\begin{aligned} \Rightarrow y_{n+s} &= \phi_{n+s} - \sum_{i=0}^{s-1} \alpha_i \left(\sum_{j=1}^{s-1} \theta_{n+i-j} \hat{y}_j + \sum_{j=s}^{n+i} \theta_{n+i-j} \hat{\phi}_j \right) \\ &= \phi_{n+s} - \sum_{i=0}^{s-1} \hat{y}_j \sum_{i=0}^{s-1} \alpha_i \theta_{n+i-j} + \sum_{j=s}^{n+i} \phi_j \sum_{i=0}^{s-1} \alpha_i \theta_{n+i-j} \\ &= \phi_{n+s} + \sum_{j=0}^{s-1} \theta_{n+s-j} \hat{y}_j + \sum_{j=s}^{n+s-1} \theta_{n+s-j} \hat{\phi}_j \\ &= \sum_{j=0}^{s-1} \theta_{n+s-j} \hat{y}_j + \sum_{j=s}^{n+s-1} \theta_{n+s-j} \hat{\phi}_j \end{aligned}$$

令 $n+s = n$ 即所欲证式。

Exercise 10.87

考虑 IVPS: $u' = f(t) = 0, u(0) = 1$, 真解为 1.

LMM 方法: $\sum_{j=0}^s \alpha_j U^{n+j} = 0$ 因为收敛, 所以初值满足 $\lim_{k \rightarrow 0} U^j = 1, j = 0, 1, \dots, s-1$, 解满足 $\lim_{k \rightarrow 0, Nk=t} U^N = 1$.

则对 $\sum_{j=0}^s \alpha_j U^{n+j} = 0$ 中 k 取极限 0, 有 $\sum_{j=0}^s \alpha_j = 0$, 则 LMM 满足 preconsistent.

考虑 IVPS: $u' = f(t) = 1, u(0) = 0$, 真解为 t .

LMM 方法: $\sum_{j=0}^s \alpha_j U^{n+j} = k \sum_{j=0}^s \beta_j$ 因为收敛, 所以初值满足 $\lim_{k \rightarrow 0} U^j = 0, j = 0, 1, \dots, s-1$, 解满足 $\lim_{k \rightarrow 0, Nk=t} U^N = t$.

则对 $\sum_{j=0}^s \alpha_j U^{n+j} = k \sum_{j=0}^s \beta_j$ 两边同时对 k 取极限 0, 因为 $Nk=t$, 有 $\sum_{j=0}^s j \alpha_j = \sum_{j=0}^s \beta_j$, 则 LMM 满足 consistent.

Exercise 10.102

Adams-Bashforth 方法

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 orders = [1, 2, 3, 4, 5]
5 theta = np.linspace(0, 2 * np.pi, 1000)
6 z = np.exp(1j * theta)
7
8 plt.figure(figsize=(10, 6))
```

```
9 plt.title("Adams-Bashforth 绝对稳定域 (p=1 到 5)")
10 plt.xlabel("Re( )")
11 plt.ylabel("Im( )")
12 plt.grid(True)
13
14 for p in orders:
15     if p == 1:
16         rho_coeff = [1, -1]
17         sigma_coeff = [0, 1]
18     elif p == 2:
19         rho_coeff = [1, -1, 0]
20         sigma_coeff = [0, 3/2, -1/2]
21     elif p == 3:
22         rho_coeff = [1, -1, 0, 0]
23         sigma_coeff = [0, 23/12, -16/12, 5/12]
24     elif p == 4:
25         rho_coeff = [1, -1, 0, 0, 0]
26         sigma_coeff = [0, 55/24, -59/24, 37/24, -9/24]
27     elif p == 5:
28         rho_coeff = [1, -1, 0, 0, 0, 0]
29         sigma_coeff = [0, 1901/720, -2774/720, 2616/720, -1274/720, 251/720]
30
31     rho = np.polyval(rho_coeff, z)
32     sigma = np.polyval(sigma_coeff, z)
33     gamma = rho / sigma
34
35     plt.plot(np.real(gamma), np.imag(gamma), label=f"p={p}")
36
37 plt.legend()
38 plt.show()
```

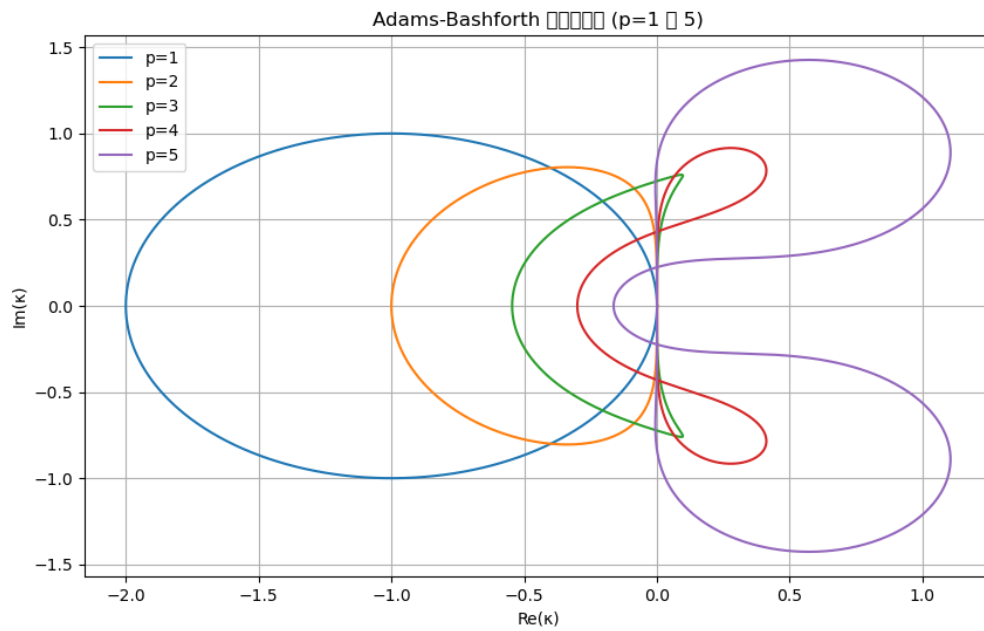


图 1: Adams-Bashforth 绝对稳定域 (p=1 到 5)

Adams-Moulton 方法

```

1  import matplotlib.pyplot as plt
2
3  orders = [3, 4, 5]
4  theta = np.linspace(0, 2 * np.pi, 1000)
5  z = np.exp(1j * theta)
6
7  plt.figure(figsize=(10, 6))
8  plt.title("Adams-Moulton 绝对稳定域 (p=3 到 5)")
9  plt.xlabel("Re( )")
10 plt.ylabel("Im( )")
11 plt.grid(True)
12
13 for p in orders:
14     if p == 3:
15         rho_coeff = [1, -1, 0, 0]
16         sigma_coeff = [5/12, 2/3, -1/12, 0]
17     elif p == 4:
18         rho_coeff = [1, -1, 0, 0, 0]
19         sigma_coeff = [3/8, 19/24, -5/24, 1/24, 0]
20     elif p == 5:
21         rho_coeff = [1, -1, 0, 0, 0, 0]
22         sigma_coeff = [251/720, 646/720, -264/720, 106/720, -19/720, 0]
23
24     rho = np.polyval(rho_coeff, z)

```

```

25     sigma = np.polyval(sigma_coeff, z)
26     gamma = rho / sigma
27
28     plt.plot(np.real(gamma), np.imag(gamma), label="p={}".format(p))
29
30 plt.legend()
31 plt.show()

```

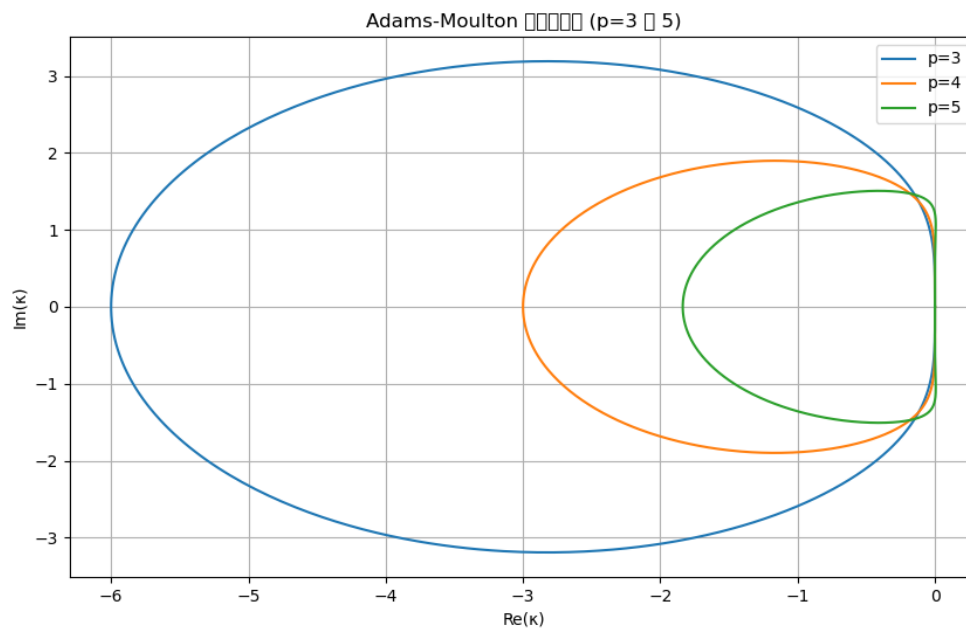


图 2: Adams-Moulton 绝对稳定域 (p=3 到 5)

BDF 方法

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  orders = [3, 4, 5]
5  theta = np.linspace(0, 2 * np.pi, 1000)
6  z = np.exp(1j * theta)
7
8  plt.figure(figsize=(10, 6))
9  plt.title("Adams-Moulton 绝对稳定域 (p=3 到 5)")
10 plt.xlabel("Re( )")
11 plt.ylabel("Im( )")
12 plt.grid(True)
13
14 for p in orders:
15     if p == 3:
16         rho_coeff = [1, -1, 0, 0]

```

```

17     sigma_coeff = [5/12, 2/3, -1/12, 0]
18 elif p == 4:
19     rho_coeff = [1, -1, 0, 0, 0]
20     sigma_coeff = [3/8, 19/24, -5/24, 1/24, 0]
21 elif p == 5:
22     rho_coeff = [1, -1, 0, 0, 0, 0]
23     sigma_coeff = [251/720, 646/720, -264/720, 106/720, -19/720, 0]
24
25 rho = np.polyval(rho_coeff, z)
26 sigma = np.polyval(sigma_coeff, z)
27 gamma = rho / sigma
28
29 plt.plot(np.real(gamma), np.imag(gamma), label="p={}".format(p))
30
31 plt.legend()
32 plt.show()

```

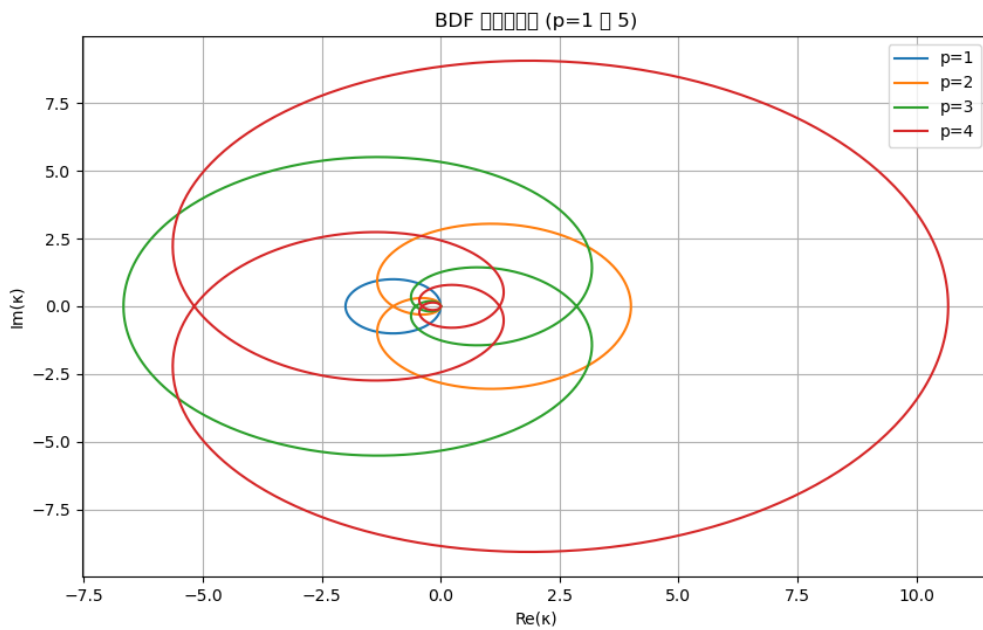


图 3: BDF 绝对稳定域 (p=3 到 5)